FIITJEE

ALL INDIA TEST SERIES

PART TEST – II

JEE (Advanced)-2020

PAPER – 1

TEST DATE: 17-11-2019

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. B, C

Sol. Magnetic flux through the smaller loop due to current in the bigger loop = $\left(\frac{\mu_0 I}{2\pi d}-\frac{\mu_0 I}{2\pi(2d)}\right)\ell^2$

$=\frac{\mu_0 I\ell^2}{4\pi d}$

So mutual inductance = $\frac{\mu_0\ell^2}{4\pi d}$

$\therefore$ So emf induced in the bigger loop = $\frac{d}{dt}\left[\frac{\mu_0\ell^2}{4\pi d}kt\right]$

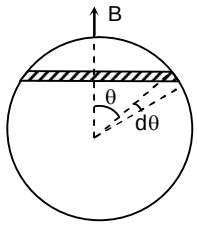
$=\frac{\mu_0\ell^2 k}{4\pi d}$

2. B, D

Sol. $\varepsilon=\pi R^2\sin^2\theta\alpha$

So current $di=\frac{\varepsilon}{dr}=\frac{1}{2\rho}dR^2\alpha\sin\theta d\theta$

So $\int d\mu=\int_0^{\pi}\frac{\pi dR^4\alpha\sin^3\theta d\theta}{2\rho}=\frac{2\pi}{3\rho}dR^4\alpha$.



3. A, B, D

Sol. Loop equation gives

$$L\frac{di}{di}+iR=\frac{\mu_0 Na^2\omega}{2\ell}i$$

$$\text{So } \frac{di}{dt}=\frac{1}{L}\left(\frac{\mu_0 Na^2\omega}{2\ell}-R\right)i$$

So to grow current always, $\frac{\mu_0 Na^2\omega}{2\ell}-R>0$

$$\Rightarrow \omega>\frac{2\ell R}{\mu_0 Na^2}$$

4. C

Sol. $eE = mr\alpha \Rightarrow E=\frac{mr\alpha}{e}$

$$\text{So } I=\frac{2\pi rE}{R}=\frac{2\pi mr^2\alpha}{eR}$$

and $B=\mu_0 nI$

$$=\frac{2\pi\mu_0 nmr^2\alpha}{eR}$$

5. B, C

Sol. During melting of ice, its volume decreases.

6. A, D

Sol. Dipole is in uniform field, so net electric force is zero.

Electric potential energy of a dipole $U=-\vec{P}.\vec{E}$. For stable equilibrium potential energy should be minimum.

7. A, C, D

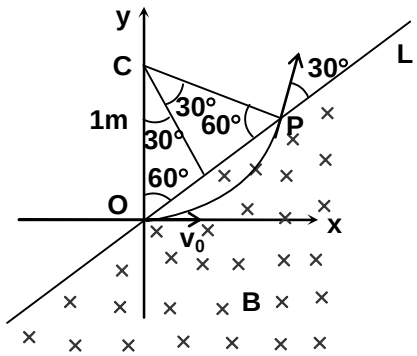
Sol. OB = 1 m

and time period

$$T=\frac{2\pi m}{qB}=2\pi \text{ sec}$$

Time spent by the particle in the magnetic field is

$$t=\frac{\pi}{3\omega}=\frac{\pi m}{3qB}=\frac{\pi}{3}\text{sec}$$



8. C

9. D

10. B

Sol. (for Q.8-10)

Cyclic process	Work Done	ΔU
(I)	$W_{AB} > 0$	$\Delta U_{AB} > 0$
	$W_{BC} = 0$	$\Delta U_{BC} < 0$
	$W_{CA} < 0$	$\Delta U_{CA} < 0$
(II)	$W_{AB} = 0$	$\Delta U_{AB} > 0$
	$W_{BC} > 0$	$\Delta U_{BC} = 0$
	$W_{CA} < 0$	$\Delta U_{CA} < 0$
(III)	$W_{AB} > 0$	$\Delta U_{AB} > 0$
	$W_{BC} > 0$	$\Delta U_{BC} < 0$
	$W_{CA} < 0$	$\Delta U_{CA} = 0$
(IV)	$W_{AB} = 0$	$\Delta U_{AB} > 0$
	$W_{BC} > 0$	$\Delta U_{BC} = 0$
	$W_{CA} < 0$	$\Delta U_{CA} < 0$

11. C

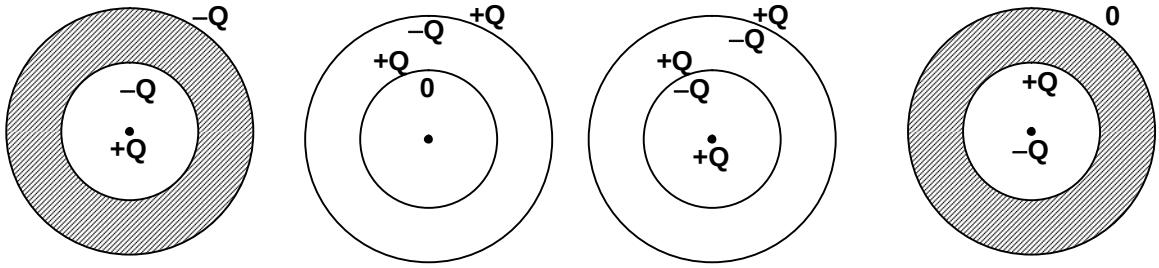
12. A

13. D

Sol. (for Q. 11-13)

Final charge distributions are as follows

Electric field at P_1 = $\dfrac{\left(9\times10^{9}\right)\left(10^{-6}\right)}{\left(10^{-1}\right)^{2}} = 900\dfrac{\text{KV}}{\text{m}}$



Electric potential of conductor = $-\dfrac{\left(9\times10^{9}\right)\left(10^{-6}\right)}{2\times10^{-1}} = -45\text{KV}$

Electric potential energy in the outside space = $\dfrac{\left(9\times10^{9}\right)\left(10^{-6}\right)^{2}}{2\times2\times10^{-1}} = 2.25\times10^{-2}\text{J}$

SECTION – B

14. 9

Sol. $KA\dfrac{dT}{dx}\bigg|_{Y} - KA\dfrac{dT}{dx}\bigg|_{X} = ms\dfrac{dT}{dt} = \left(10^{-6}\times8\times10^{3}\right)(500)(0.05)$

$$\therefore \quad \left.\frac{dT}{dx}\right|_Y = \left.\frac{dT}{dx}\right|_X + \frac{10^{-6}\times 8\times 10^3\times 500\times 0.05}{500\times 10^{-4}} = 9$$

15. 5

Sol. $$\phi = \frac{q}{24\varepsilon_0} - \frac{q}{\varepsilon_0 4\pi}\frac{2\pi\left(1-\frac{1}{\sqrt{2}}\right)}{4}$$

$$= \frac{q}{24\varepsilon_0} - \frac{q}{8\varepsilon_0} + \frac{q}{8\sqrt{2}\varepsilon_0} = \frac{q}{8\sqrt{2}\varepsilon_0} - \frac{q}{12\varepsilon_0}$$

$$\approx 5 \ \frac{Nm^2}{c}.$$

16. 4

Sol. $R_S = 7\times 10^{-3} = \frac{\rho_S \ell_S}{\pi r_S^2}$ (X_S is the physical quantity in space)

So $7\times 10^{-3} = \frac{\rho_\ell (1-\beta 300)\,\ell_\ell (1-\alpha 300)}{\pi r_\ell^2 (1-\alpha 300)^2}$

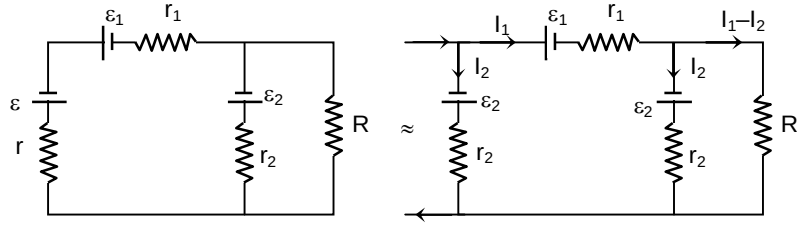
Where (X_ℓ is the physical quantity in ISRO lab)

Putting all the values

$\ell_\ell \approx \frac{1}{4}$m

17. 5

Sol. For the cell network equivalent emf is ε and equivalent resistance is r, then the equivalent circuit is



So loop equations is the left loop is

$-1-4I_1+2-6I_2+6I_2-2=0$

So $I_1 = -\frac{1}{4}$A

Another loop equation is the right loop is

$-(I_1-I_2)+6I_2-2=0$

So $7I_2 = 2-\frac{1}{4}$ So $I_2 = \frac{1}{4}$A

So $I_1 - I_2 = -\frac{1}{2}$A

18. 9

Sol. $F = \int_0^R \frac{Q}{\pi R^2} 2\pi r dr\ \omega r B_0 \Rightarrow \frac{2}{3} Q\omega B_0 R = mg$

$\omega = 9\times 10^2$ rad / s

Chemistry

PART – II

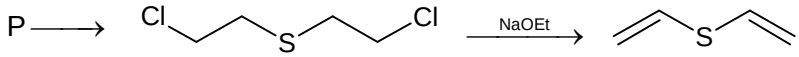
SECTION – A

19. B, D
Sol. Pyrrole is π-excessive heterocyclic compound, so can take part in Reimer-Tiemann reaction.

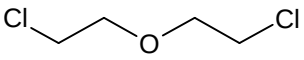
20. B, C, D
Sol. Cellulose nitrate is a semi-synthetic polymer.

21. A, C, D
Sol. Conceptual

22. A, B, C
Sol. Allyl and benzyl 1° alcohol give turbidity with Lucas reagent instantly.

23. A, B, C, D
Sol. 

(Mustard gas)
Rate of hydrolysis is more than corresponding ether analogue



due to NGP.

24. A, B, C
Sol. Bridgehead β-keto carboxylic acids do not decarboxylate as it would require formation of a double bond at bridgehead carbon which would be highly strained.

25. A, B, C, D
Sol. Epimers, Anomers and geometrical isomers – all are examples of diastereomers.

26. B

27. A

28. D
Sol. (for the Q. No. 26 to 28)
Factual.

29. B

30. A

31. B
Sol. (for the Q. No. 29 to 31)
Conceptual.

SECTION – B

32. 3

Sol. For A $\Rightarrow 2^6$

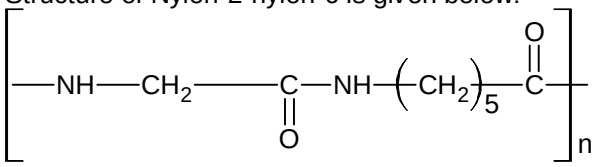
For B $\Rightarrow 2^2$

33. 9

Sol. $A \longrightarrow CH_3 - \overset{O}{\overset{\|}{C}} - \overset{O}{\overset{\|}{C}} - OH$

34. 6

Sol. Structure of Nylon-2-nylon-6 is given below:



35. 6

Sol. Nitrobenzene is highly deactivating towards EAS reactions, cannot take part in Friedel-Crafts alkylation.

Aryl 1° amine cannot be prepared by Gabriel's phthalimide synthesis.

Aniline give very poor yield in Friedel-Crafts alkylation.

Phenol on treatment with bromine water gives tri-substituted product.

36. 4

Sol. Conceptual.

Mathematics

PART – III

SECTION – A

37. B, D

Sol. $\tan^{-1} x - \tan^{-1} y = \tan^{-1}\frac{1}{8}$

Case-I: $x, y > 0 \Rightarrow \frac{x-y}{1+xy} = \frac{1}{8} \Rightarrow x = \frac{8y+1}{8-y} \Rightarrow$ (5, 3) and (57, 7)

Case-II: $x > 0, y < 0 \Rightarrow \frac{x+p}{1-xp} = \frac{1}{8}$ $(p = -y) \Rightarrow x = \frac{1-8p}{8+p} < 0 \ \forall\ p > 0 \Rightarrow$ no solution

Case-III: $x < 0, y < 0 \Rightarrow \tan^{-1} p - \tan^{-1} q = \tan^{-1}\frac{1}{8}$ $(p = -y, q = -x) \Rightarrow (-3, -5), (-7, -57)$

38. A, B, C

Sol. $E_c : x^2 = cy$

$F : (y^2 - x)(y^2 + x) = 0$

39. A, C, D

Sol. Bisectors are given by $\frac{3x-4y}{5} = \pm\left(\frac{x+y-4}{\sqrt{2}}\right)$. Bisector containing (2, 1) is given by '–' sign which is also obtuse angle bisector

40. A, B, D

Sol. Tangent for ellipse : $y = mx \pm \sqrt{m^2+2}$

Tangent for hyperbola : $y = mx \pm \sqrt{2m^2-2}$

$\Rightarrow m^2 = 4$ or $m = \pm 2$

Point of tangency for ellipse $\left(\frac{-2}{\sqrt{6}}, \frac{2}{\sqrt{6}}\right)$ and for $\left(\frac{-4}{\sqrt{6}}, \frac{-2}{\sqrt{6}}\right)$

41. A, C, D

Sol. The two ellipses will be reflection of each other in common tangent

So, locus of centre of E_2 can be obtained by finding locus of centre of E_1 in variable tangent to E_1.

If (h, k) be the reflection of centre, then $\frac{h-0}{m} = \frac{k-0}{-1} = -2\left(\frac{\sqrt{4m^2+3}}{m^2+1}\right)$

Eliminating 'm', we get $(x^2 + y^2)^2 = 4(4x^2 + 3y^2)$ which is fourth degree polynomial closed curve

Similar procedure can be adopted for focus

42. A, B

Sol. $\Delta_1 = 2R^2 \sin A \cos B \cos C$

$\Delta_2 = 2R^2 \cos A \sin B \cos C$

$\Delta_3 = 2R^2 \cos A \cos B \sin C$

$\Delta_1 + \Delta_2 + \Delta_3 = 2R^2(\sin(A + B + C) + \sin A \sin B \sin C) = 2R^2 \sin A \sin B \sin C = \Delta$

$R_1 = R_2 = R_3 = R$

43. B, C

Sol. $2019 = 3 \times 673$

$2^{a_1 \cos^2 a_1}\ 3^{a_2 \cos^2 a_2}\ 5^{a_3 \cos^2 a_3} \ldots\ldots = 3 \times 673$

$\Rightarrow a_1 = a_3 ..... = 0,\ a_2 \cos^2 a_2 = 1$ and $a_{122} \cos^2 a_{122} = 1$ ($a_{122} = 673$)

$x \cos^2 x = 1$ has 5 solutions in $(-\infty, 9]$, infinite solutions in $(-\infty, \infty)$, no solution in $(-\infty, 0]$

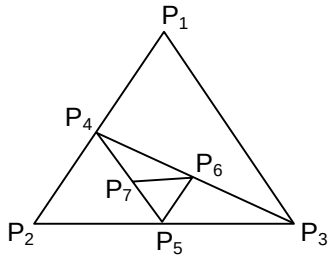
44. C

45. A

46. B

Sol. (for Q. 44.-46.)

If $P_1 P_2 P_3$ is equilateral then $P_5 P_6 P_7$ is also equilateral



47. C

48. A

49. A

Sol. (for Q. 47.-49.)

The identities are $\sin^{-1} x - \cos^{-1} x = \frac{\pi}{2}\ \forall\ x \in [-1, 1]$

$\cot^{-1} x + \tan^{-1} x = \frac{5\pi}{2}\ \forall\ x$

$\sec^{-1} x + \text{cosec}^{-1} x = \frac{\pi}{2}\ \forall\ x \in R - (-1, 1)$

$\sin^{-1} x + \sin^{-1}(-x) = 2\pi\ \forall\ x \in [-1, 1]$

$\tan^{-1}(-x) + \tan^{-1} x = 2\pi\ \forall\ x \in R$

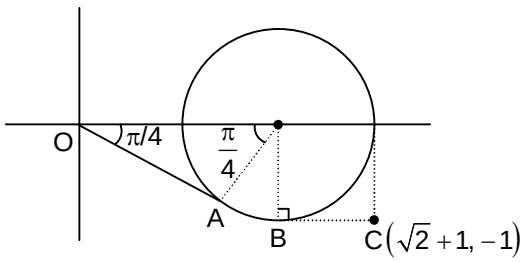
$\cot^{-1}(-x) + \cot^{-1} x = 3\pi\ \forall\ x \in R$

$\text{cosec}^{-1}\left(\frac{1}{x}\right) = \pi - \sin^{-1} x\ \forall\ x \in [-1, 1] - \{0\}$

SECTION – B

50. 2

Sol. Length of path OABC is $\frac{\pi}{4} + 2$



51. 8

Sol. f(a, b) represent distance between the curves $16x^2 + 9y^2 = 144$ and $x^2 + y^2 = 25$.

So, c = 9 and d = 1

52. 4

Sol. Director circle of $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is $x^2 + y^2 = a^2 + b^2$. It intersects $(x - 1)^2 - y^2 + \frac{19}{2}=0$ at only two points. Solving the two equations simultaneously $2x^2 - 2x + \frac{21}{2} - (a^2 + b^2) = 0$

$\Rightarrow 4-8\left(\frac{21}{2}-\left(a^2+b^2\right)\right)=0 \Rightarrow a^2 + b^2 = 10 \Rightarrow a + b = 4$

53. 8

Sol. $\text{cosec}^2 x - \cot^2 x - (1 + \tan^2 x) - \tan^2 x - 1 + 3 = 0$

$\Rightarrow \tan^2 x = 1 \Rightarrow \tan x = \pm 1$

54. 3

Sol. Domain of expression $f(x)=\tan^{-1}x+2\sin^{-1}x+3\sec^{-1}x$ is $\{-1,1\}$

$f(-1)=-\frac{\pi}{4}+2\left(-\frac{\pi}{2}\right)+3\pi=\frac{7\pi}{4}$

$f(1)=\frac{\pi}{4}+2\left(\frac{\pi}{2}\right)+3.0=\frac{5\pi}{4}$